

Project Ariel: An Open Source Augmented Reality Headset for Industrial Applications

James Campbell*

campbellfabrications@gmail.com

Brewed Engagement Extended Reality Labs
Adelaide, South Australia, Australia

Alvaro Cassinelli

cassinelli.alvaro@gmail.com

School of Creative Media
Hong Kong City University
Kowloon, Hong Kong

Vincent Ta

vincentta911@gmail.com

Brewed Engagement Extended Reality Labs
Adelaide, South Australia, Australia

Damien Constantine Rompapas*

dr-damo@beer-labs.net

Brewed Engagement Extended Reality Labs
Adelaide, South Australia, Australia

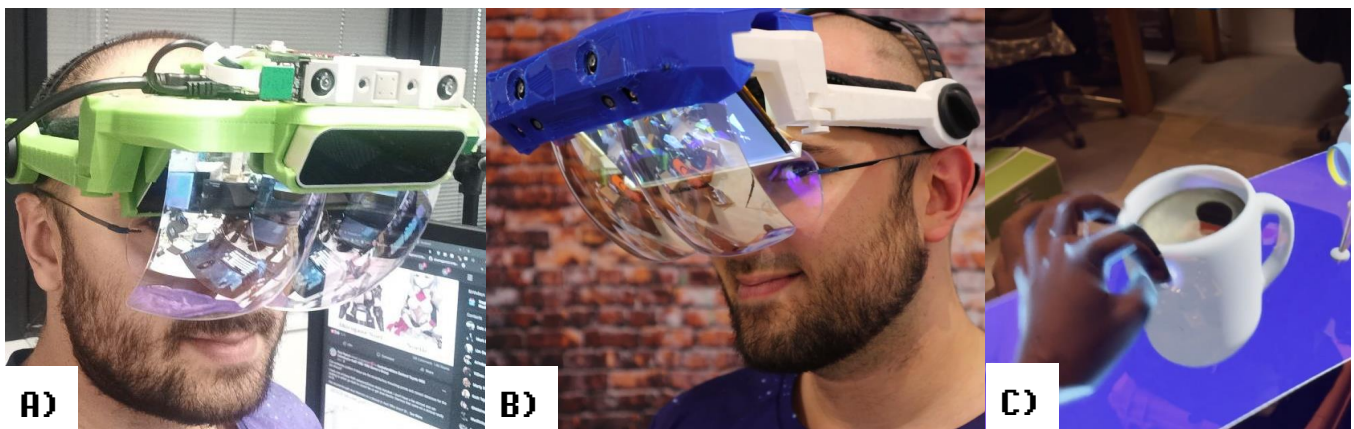


Figure 1: We introduce Project Ariel, an open source Augmented Reality modular headset specifically designed for use in an industrial setting. A) The original Ariel design, prioritizing cheaper components for scalability purposes, at the cost of additional weight. B) The second version of Ariel, slightly increasing the cost of the build while greatly reducing bulk and weight. C) An image captured through the headset after optical and sensor calibration, showcasing the visual quality achievable by Ariel.

ABSTRACT

Some of the biggest challenges in applying Augmented Reality (AR) technologies to the industry floor are in the form factor, and safety requirements of the head worn display. This includes alleviating issues such as peripheral view occlusion, and adaptation to personal protective equipment. In this work we present the design of Project Ariel, an Open Source 3D printable display specifically designed for use in industrial environments. It is our hope that with this technology, the average tradesman can utilize the powerful

*Both authors contributed equally to this research.

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UbiComp-ISWC '21 Adjunct, September 21–26, 2021, Virtual, USA
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ACM ISBN 978-1-4503-8461-2/21/09...\$15.00
<https://doi.org/10.1145/3460418.3479359>

visualizations AR has to offer, significantly improving their daily work flow.

KEYWORDS

Augmented Reality; Headset Design; Optical See-Through; Open Source;

ACM Reference Format:

James Campbell, Vincent Ta, Alvaro Cassinelli, and Damien Constantine Rompapas. 2021. Project Ariel: An Open Source Augmented Reality Headset for Industrial Applications. In *Adjunct Proceedings of the 2021 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers (UbiComp-ISWC '21 Adjunct)*, September 21–26, 2021, Virtual, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3460418.3479359>

1 INTRODUCTION

A working tradesman's dream is the ability to take in information about the task being performed without exerting any mental effort, allowing him to better focus on the manual task at hand.

Augmented Reality (AR), usually deployed with head worn display (HWD), is a technology that embeds Computer Graphics (CG) into the user's view of the environment, perfect for delivering visual information about the real world to the wearer. Current industry standard headsets, such as the Microsoft HoloLens, or Magic Leap One, are such HWDs that can be applied in the situation. However, the devices are expensive, making mass adoption in the field near impossible for the average working tradesmen. Furthermore, they do not fit the International Organization for Standards (ISO) as the peripheral view is obstructed by the optics frame. Finally, Personal Protective Equipment (PPE) must be modified to adapt to the headsets, potentially voiding the protection they offer.

Project North Star[6] is an open source, tethered HWD, with a wide viewing angle. The 3D printed design allows for modification to suit fitting PPEs, without modifying the PPE. However, the placement of the displays block the horizontal peripheral view of the wearer. Project Triton[1] is an alternative 3D printed design with an ideal form factor, placing the display in front of the user's forehead, only blocking the upper peripheral view. However, it lacks any head tracking solution, crucial for accurately placing AR content within the environment[5].

In this work, we introduce the early work designs of Project Ariel, an Open Source AR HWD inspired from the Project Triton form factor. By placing the display in front of the forehead of the user, and designing the HWD's frame to allow for easily adapting to PPE, this HWD aims to become a framework for custom-fitting to PPEs, applying AR technologies to previously un-reachable tradesmen workplaces.

2 REQUIREMENTS FOR DEPLOYING AUGMENTED REALITY IN AN INDUSTRIAL ENVIRONMENT

In order for us to apply AR technologies in the industrial space, we interviewed several individuals from various industrial trades, this includes welding/fabrication, tank inspection, automotive mechanical, and robotics/assembly line workers. We also interviewed several people within the Agricultural department, both produce sorting and field workers. From the discussions, and following the International Organization for Standards (ISO) we derived the following minimum criteria that must be met:

- The headset must not block the horizontal peripheral view.
- When inactive, the headset must not obstruct the user's view of the real world.
- The front of the headset must be modular, to allow attachment of facial protective equipment (Such as a welding mask, or face shield).
- The AR content must not require an additional peripheral, limiting interactions by the user's hands.
- The headset must have a means to either quickly doff, or flip up to allow rapid deactivation/removal of the occluding AR content.
- The headset must be as light as possible to prevent fatigue, as it will be worn during an entire work day.
- The headset may be tethered to a backpack, belt-worn, or side carried portable compute solution. There is a preference

of belt-worn compute pc's, since most tradesmen already carry belt holstered tools.

3 DESIGNING PROJECT ARIEL - DETERMINING THE HARDWARE

Analyzing the available open source headsets, Project Triton is the closest fit to our end target. This is due to the placement of the screen allowing for an unblocked horizontal peripheral field of view, as opposed to the Project NorthStar headset. Additionally, the open source design of Project Triton is modular, allowing us to modify the frame to fit with PPEs. However, the current form factor of Project Triton does not facilitate the mounting of a head tracking unit, and the optical setup exhibits small amounts of blur at the bottom of the screen. Finally, there is no optical or between sensor calibration setup specific to Project Triton's optics, meaning any image displayed through the headset will exhibit some distortion. In this section we briefly go over possible options for head tracking, and hand tracking, describing what solution was used, and why.

One of the core requirements for AR content to be placed within the environment, is that the device has some means of tracking the user's head within the environment. Rompapas Et Al[5] describes three typical ways to achieve this:

- Fiducial marker tracking, where sensors placed on the device track fiducial markers located within the environment
- Markerless Inside-Out tracking, where sensors placed on the device track features in the environment
- Outside-In tracking, where external sensors are placed in the environment and track markers placed on the head worn device.

Since our target is industrial applications, it's unreasonable to assume that the environment can be modified to place external outside-in tracking sensors or fiducial markers. Because of this, we focus entirely on Markerless Inside-Out tracking sensors. Figure 2 shows a summary of possible off the shelf sensors available. The Stereo Labs sensor series, along with several open source Simultaneous Localization And Mapping (SLAM) solutions compute the head tracking on the device the sensor is connected to. While acceptable if being run with higher compute power, it uses up valuable resource space which may not be available for smaller and cheaper compute pack hardware. This reasoning is why we favour the Intel Realsense T265 (Standalone) and T261 (Sensor only without casing) tracking hardware. The realsense tracking device performs SLAM on chip, saving valuable resources.

Due to the industrial requirements preventing us from adding peripherals to the tradesmans equipment, we decided to use the ultraleap leapmotion controller that already fits to the Project Triton headset. We also created a variation of the headset to remove as much weight as possible, using the ultraleap rigel sensor device without it's aluminium casing.

In order to alleviate the limitations of project triton's form factor, we performed the following modifications to the 3D printed files:

- Tilting the optics bracket towards the display around 10 degrees, to bring the full display into focus
- Moving the display further down to fill the optical combiner area

	Zed (Stereo Labs)	Zed Mini (Stereo Labs)	Zed 2 (Stereo Labs)	T261/5 (Intel)	Structure Core (Occipital)
Price (USD)	349	399	499	150	400
SLAM Rate (Hz)	100	100	100	30	60
IMU Rate (Hz)	Nil	400	800	200	1000
Spatial Reconstruction	Y	Y	Y	N	Y
Restriction	NVIDIA GPU	NVIDIA GPU	NVIDIA GPU	None	None
Processing Location	Host	Host	Host	Chip	Host
Weight (g)	135	62	166	22	26

Figure 2: Our comparison between several off-the-shelf 6DOF tracking units that can serve as a viable pose feed for head tracking. While the T261/5 series is missing spatial reconstruction, it has no restriction on what pc hardware can operate it, nor does it perform the SLAM processing on the host, saving valuable resources in low-cost situations. For these reasons, we selected the T261/5 to use in Ariel.

- Carving into the front of the headset to allow fitting of a sensor mount bracket, to place the t261 and the leap motion tracker/rigel on the front of the headset
- Modifying the ear piece mounts to accommodate mounting to a standard welding helmet harness, allowing for the same flip up access a standard welding helmet allows.
- Creating a fully housed cover, that can be modified further to fit to PPE equipment, and protects the exposed sensors.

4 ASSEMBLING AND SETTING UP PROJECT ARIEL

Assembly of the Project Ariel headset is straight forward, Figure 3 shows an exploded diagram showing how the headset is assembled. For a complete Bill of Materials, please see the online resources section at the end of this paper. After 3D printing the files, the build is ordered as follows:

- (1) First, the display is slid into the display bracket
- (2) The display driver boards are bolted onto the display bracket, the display ribbon cable is folded and inserted through the loop on the side of the display bracket, then clamped onto the driver breakout board.
- (3) The breakout board's MIPI cable is looped through the top of the display bracket, to the top of the headset, to connect to the MIPI display driver.
- (4) The MIPI display driver is secured using the standoff part, to the top of the display driver board.
- (5) The integrator electronic component is bolted to the top of the standoff part.
- (6) The leapmotion/rigel is bolted to the sensor mount bracket, which is then bolted to the front of the headset, all sensor cables are connected to the integrator electronic component
- (7) Finally, the lid of the headset slides over the front, and bolts to the top of the headset. The earpieces for the welding helmet/head strap harness are slid into the ear marks, secured by the 3D printed bolts.

Overall, the headset takes around two hours to assemble. The total cost for the build is roughly 350 USD if using the leap motion controller, and 400 USD if using the ultraleap rigel hand tracking sensor. The headset weights 480 grams total, making it lighter than the Microsoft HoloLens 2, although most of the weight is placed on the front of the headset, which is not ideal.

4.1 Calibrating the Optics

Before the headset can be used, the optics of the headset need to be calibrated so that all lines appear straight when viewed through the headset. For this we use the same approach described in Project Esky [2], re-describing it for completeness. We place a single stereo camera pair (In our case use the t261), placed behind the view of the

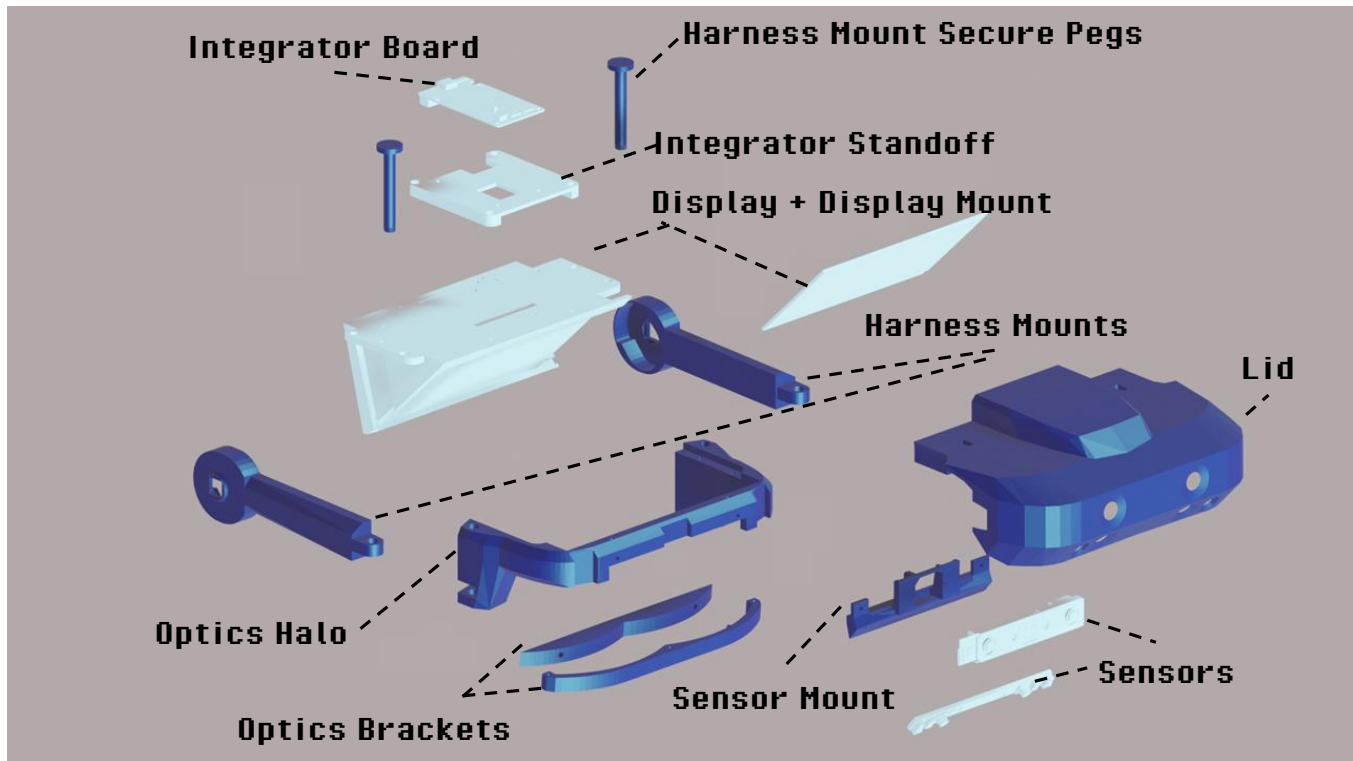


Figure 3: An exploded view of the 3D parts and electronics that the Ariel headset is comprised of.

headset. We then use binary-coded structured light homography [3] to compute the depth of each pixel displayed on the HMD, in order to collect a series of undistortion points. These points allow us to create a look up table, resolving a 3rd-order 2D polynomial from the 2D screen space to the corresponding point on the now undistorted virtual viewport. As mentioned by Rompapas Et. Al, this calibration technique is not perfect, and exhibits distortion at the extreme edges of the display. It also limits the Inter Pupillary distance of the viewer to the stereo camera’s between sensor distance (typically 64mm). Overall, the procedure for calibrating the optics takes roughly five minutes to perform, and is not required to be performed for multiple users, although ideal to do so provided there is a means for removing the IPD limitation.

4.2 Calibrating the Sensor Offsets

Before the 6DOF and hand tracking can be utilized with our now-calibrated HMD, their offsets from the user’s viewport need to be calculated. We use either of the two approaches listed in Project Esky [2]. Again describing them for completeness.

Online Hand-based Alignment: A user holds their hand in front of the leap motion tracker, which shows the virtual hand with some offset, then the user presses a button, freezing the virtual hand in place, allowing the user to match the virtual hand’s pose and orientation with their real hand. Doing this several times captures the corresponding points between the initial fingertip pose and orientation. Then by using a simplified Kabsch Solve method to align the two point sets, we can resolve the transform between

the hand tracker and the user’s eye, as well as the 6DOF tracker with respect to the hand tracker. While this method does work, and requires less hardware to perform, it can lead to errors generated by user input. This process takes roughly 10 seconds, and does not need to be repeated between users.

Offline Visual Marker Alignment: This process involves a stereo camera placed behind the lens of the OST-HMD, with an ARuCO marker placed in the view of all sensors. The relative pose between each sensor is computed automatically by inferring the transform between each sensor and the detected ARuCO marker. This technique involves the use of extra hardware, but allows for an automated calculation of the required pose offsets. This also has the benefit of calculating where the user’s eyes are expected to be placed behind the headset (therefore allowing us to place the stereo rendering cameras where the user’s eyes would be in virtual space). This procedure takes roughly 10 seconds, and does not need to be repeated between users.

4.3 Deployed hardware and software setup

Once the optics and sensor offsets are calibrated, we are now free to develop applications specifically for use in the field. Our headset is a tethered AR headset, meaning that it can be powered by a PC or compute pack. In our case, we use an AMD Ryzen Embedded V-Series V1605B Compute Pack, featuring a Radeon Vega 8 Graphics chipset, 2 GHz 4 core CPU, and 4GB RAM. This is powered with a 28,000 MaH 100W power delivery capable battery bank, allowing for up to 2 hours of usage time. For software and development, we

use the unity game engine. More specifically, we use the Project Esky software framework[2], including Microsoft's Mixed Reality toolkit, and asynchronous rendering pipeline to deploy full AR functionality. Alternative software setups include the Monado open source runtime [4], which includes the 2D optical undistortion technique for rendering described in this paper.

5 CONCLUSION

In this paper, we present Project Ariel, an open source 3D printed headset specifically designed to address the shortcomings of other open source 3D printed headsets in industrial settings, making it possible to be deployed in industrial settings.

There are several limitations that need to be addressed in the design of the headset to improve the visual quality, even if it's usable in its current state. The cheaper LCD panels used in the headset have a high latency ($> 30\text{ms}$), and a slow refresh rate (50Hz). While this does limit the fidelity of the AR content, the device is still capable of accurately embedding AR content into the real world. Additionally, some minor bleed occurs through the headset from internal reflection, a re-design of the optics shape can alleviate this, or one can switch out the 6 inch display to a smaller screen size, sacrificing the AR field of view.

Our intention is to deploy this in industrial settings. One example of this is in welding, by marking a virtual line along the edge of where the weld needs to be made, the viewer is able to see this visual guide despite the welding helmet darkening the user's view of the real world. Another application lies in fruit picking, one could use the exposed sensor images to detect fruit that lies in the tree, grading it. One could take this a step further, implementing this as a video game where pickers earn 'bonus points' upon picking well graded fruit.

ACKNOWLEDGMENTS

The authors would like to thank Professor Daniel Saakes for his invaluable industrial design insights.

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A ONLINE RESOURCES

Here is listed a series of online resources for support, and building of project Ariel.

Project Ariel 3D printed parts (Including Bill of Materials):
<https://github.com/Dr-Damo/Project-Ariel>

BEER Labs Discord (For support):

<https://discord.gg/yUW69n9x3C>

Supplemental Video (Online): <https://youtu.be/uqVO9kf6VJc>